All this seems to follow readily enough from the *Voyager* observations. The rate at which the surface is covered over by current volcanic flows implies massive changes in 50 or 100 years, a prediction that luckily can be tested. The *Voyager* images of Io can be compared with much poorer images taken by ground-based telescopes 50 years earlier, and by the Hubble Space Telescope 13 years later. The surprising conclusion seems to be that the big surface markings on Io have hardly changed at all. Clearly, we're missing something.

A VOLCANO in one sense represents the insides of a planet gushing out, a wound that eventually heals itself by cooling, only to be replaced by new stigmata. Different worlds have different insides. The discovery of liquid-sulfur vulcanism on Io was a little like finding that an old acquaintance, when cut, bleeds green. You had no idea such differences were possible. He seemed so ordinary.

We are naturally eager to find additional signs of vulcanism on other worlds. On Europa, the second of the Galilean moons of Jupiter and Io's neighbor, there are no volcanic mountains at all; but molten ice—liquid water—seems to have gushed to the surface through an enormous number of crisscrossing dark markings before freezing. And further out, among the moons of Saturn, there are signs that liquid water has gushed up from the interior and wiped away impact craters. Still, we have never seen anything that might plausibly be an ice volcano in either the Jupiter or Saturn systems. On Triton, we may have observed nitrogen or methane vulcanism.

The volcanos of other worlds provide a stirring spectacle. They enhance our sense of wonder, our joy in the beauty and diversity of the Cosmos. But these exotic volcanos perform another service as well: They help us to know the volcanos of our own world—and perhaps will help one day even to predict their eruptions. If we cannot understand what's happening in other circumstances, where the physical parameters are different, how deep can our understanding be of the circumstance of most concern to us? A general theory of vulcanism must cover all cases. When we stumble